

GridGuard

AI-Powered Predictive Maintenance for Electrical Grids

PREVENT FAILURES. PRIORITIZE MAINTENANCE. PROTECT THE GRID.

HAZARD COST

\$1M+ / HOUR

Unplanned power outages cause massive economic damage to utilities & hospitals.

INTELLIGENCE ENGINE

0–100 RISK SCORE

Fuses sensor telemetry, Open-Meteo weather, incident history & wear state.

ACTIONABLE OUTCOME

**PRE-POSITION
CREWS**

Ranks high-consequence assets to stage maintenance before failure strikes.

The Problem: Why Grid Maintenance Needs a Smarter Approach

1. SILENT ASSET DETERIORATION

Power transformers & substation equipment decay internally long before visible physical breakdown occurs.

2. THE CALENDAR MAINTENANCE TRAP

Utilities rely on fixed periodic schedules, ignoring sensor signals that show failure signatures weeks in advance.

3. CONTROL ROOM DATA OVERWHELM

Operators manually monitor temperature, vibration, oil quality & partial discharge across hundreds of assets.

4. HIGH-COST UNPLANNED OUTAGES

A single equipment failure triggers widespread blackouts, costing utilities upwards of \$1M+ per hour.

CORE OPERATOR NEED: Which high-consequence asset requires attention right now before an outage occurs?

Our Solution: 0–100 Deterministic Risk Engine

EXACT RISK FORMULA (100% AUDITABLE)

overall =

(Sensor Health × 0.30) +

(Weather Risk × 0.20) +

(Failure History × 0.15) +

(Degradation × 0.15) +

(Grid Impact × 0.20)

*Independently auditable component scores. Dominant factor surfaced automatically.

⚡ Sensor Health (30%)

IEC-standard temp (°C), vibration (mm/s), oil quality, partial discharge (pC).

☁ Weather Risk (20%)

Open-Meteo rain, ambient temp stress, wind/storm severity forecasts.

📊 Historical Failures (15%)

Logged incident frequency, historical failure causes, and fault patterns.

🔧 Asset Degradation (15%)

Physical age, lifecycle state machine tracking, and cumulative fault stress.

🏠 Grid Importance (20%)

Customers served, critical facility dependencies (hospitals), cascade risk.

NORMAL (0–39)

Routine Monitoring

WATCH (40–69)

Schedule Inspection

HIGH (70–84)

Prioritize Maintenance

CRITICAL (85–100)

Pre-Position Crews Today

How GridGuard Works: End-to-End Pipeline

1. DATA INGESTION

Telemetry, Open-Meteo, Incidents & Asset Logs

2. NORMALISATION

Physical Units → 0–100 Scores via IEC Standards

3. RISK ENGINE

5-Factor Composite Deterministic Scoring

4. PRIORITISATION

Risk + Grid Impact Ranking & Crew Pre-Positioning

5. AI BRIEFING

Grounded Explanations & Operator Control UI

GridGuard Control RoomLIVE ENGINE ONLINE

8 Assets Scored

Healthy (4)

Monitoring (2)

High Risk (1)

Critical (1)

Attention Queue (Ranked by Risk & Grid Impact)

#1	Substation TX-007	North Zone	Score: 94.9	Weather (100/100)	PRE-POSITION CREW TODAY
#2	Transformer TR-104	East Zone	Score: 78.2	Sensor (88/100)	Schedule Inspection
#3	Feeder FD-302	West Zone	Score: 58.4	Degradation (65/100)	Monitor Baseline

PURE-FUNCTION STATE MACHINE

States: ACTIVE → FAULTED → RETIRED

Replacement Lineage:

Retires the old unit with full incident history intact, spawning a clean-state successor linked by predecessor_asset_id.

Zero-Dependency Engine:

Python stdlib runtime — 0 external packages or network dependencies required.

OPERATOR CONTROL LOOP: DETECT → PRIORITIZE → EXPLAIN → ACT

Physical Telemetry & IEC Normalisation Mapping

Parameter	Physical Unit	IEC Standard	Critical Threshold	Normalized Risk
Top-Oil / Winding Temp	°C (Celsius)	IEC 60599	> 95 °C Winding Stress	100 / 100 (Critical)
Partial Discharge (PD)	pC (Picocoulombs)	IEC 60270	> 500 pC Insulation Breakdown	92 / 100 (Critical)
Vibration Velocity	mm/s (RMS)	ISO 10816	> 4.5 mm/s Mechanical Stress	80 / 100 (High)
Inbound Storm Wind	km/h (Peak)	Open-Meteo	> 90 km/h Severe Front	100 / 100 (Critical)

IEC STANDARDS & SQLITE RISK SNAPSHOTS

- Normalisation converts raw engineering units into standardized 0–100 risk values calibrated against international utility thresholds.
- All score computations generate immutable SQLite snapshots, preserving full historical lineage across asset repairs and replacements.

Control Room Dashboard & Grounded AI Briefing

GridGuard Control Room

LIVE ENGINE ONLINE

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#3	Feeder FD-302	West Zone	Score: 58.4	Degradation (65/100)	Monitor Baseline
#4	Substation TX-001	Central	Score: 22.1	Normal Ops	Routine Maintenance

Guard AI — Live Assistant

Grounded Context

User: What's wrong with TX-007?

Guard AI:

TX-007 is Critical at 94.9. Main driver is weather risk (100/100) — severe storm inbound & 3 prior weather failures. Inspect today.

User: Why is weather such a big deal?

Guard AI:

Wind/storm score is 100. TX-007 has had 3 of its 4 failures triggered by weather events. The storm compounds an already degraded asset.

Recommendation: Pre-position crew before the front arrives.

GROUNDING RULE: LLM NEVER calculates risk scores — deterministic engine scores first, passing JSON context to AI.

Interactive Live Simulation: Real-Time Grid Stress-Testing

1-CLICK SIMULATE BUTTON

Integrated directly into the Maintenance Plan UI. Allows operators to trigger real-time storm stress & load spike simulations across all regional grid zones.

DYNAMIC STORM STRESS INJECTION

Injects Open-Meteo storm fronts (100/100 storm severity, precipitation, ambient heat) directly into live asset wear states before actual weather arrives.

REAL-TIME RISK SURGE

Recalculates risk scores on the fly—causing vulnerable assets like Substation TX-007 to surge to Critical (94.9) and re-ranking the Attention Queue.

CREW PRE-POSITIONING

Validates crew dispatch priorities before severe weather strikes, enabling proactive pre-positioning rather than post-blackout firefighting.

GROUNDING AI INTEGRATION

Feeds simulation state directly into Guard AI, allowing operators to ask "What happens if the storm hits TX-007?" and receive grounded reports.

SIMULATION RULE: Dynamic stress testing evaluates environmental load in-memory without altering historical SQLite asset records.

What Makes GridGuard Different? (SCADA vs GridGuard)

TRADITIONAL SCADA MONITORING (REACTIVE)

Alert: "Substation TX-007 parameter out of bounds."

Result: Operators get generic warnings without context, forcing manual data analysis while equipment decays.

GRIDGUARD INTELLIGENCE (PROACTIVE)

Actionable Path: "TX-007 Critical (94.9) → Dominant Driver: Inbound Storm → Grid Rank #1 → Pre-Position Crew Today."

Result: Complete decision loop from failure prediction to crew dispatch.

▮ Risk-Based Maintenance

Dynamic risk replacing rigid calendars.

⚡ Earlier Failure Detection

Identifies decay weeks before blackouts.

📊 Data-Driven Prioritisation

Ranks failure risk against grid impact.

▮ Grounded AI Explanations

100% auditable explanations without hallucination.

🌩️ Weather-Aware Scoring

Forecasts how storms compound physical wear.

▮ Smarter Crew Allocation

Stages repair teams where blackouts cost most.

Technical Validation & Enterprise Roadmap

546 TESTS PASSING (ZERO NETWORK CALLS)

- **test_ai_briefing.py (145 tests)**
Context grounding, prompts, OpenAI mock
- **test_normalisation.py (110 tests)**
IEC unit conversion, threshold validation
- **test_lifecycle.py (94 tests)**
State machine & predecessor lineage
- **test_storage.py (67 tests)**
SQLite schema, seeder & risk snapshots
- **test_risk_engine.py (61 tests)**
5 component scoring & determinism
- **test_weather.py (61 tests)**
Open-Meteo client & storm classifier

FUTURE ENTERPRISE ROADMAP

- ▢ **Real-Time SCADA & DNP3 Ingestion**
Connect directly to live utility telemetry & substation networks.
- ▢ **Machine Learning Failure Models**
Train deep predictive failure models on massive historical utility logs.
- ▢ **Mobile Alerts for Field Teams**
Push high-risk alerts & AI briefings directly to field technicians.
- ▢ **Work-Order System Automation**
Automatically trigger work orders in SAP / Enterprise Asset Management.
- ▢ **Digital-Twin Cascading Simulation**
Simulate load stress & storm fronts across interactive digital twins.

Conclusion: Transforming Grid Reliability

Predict before you react. Prioritize before you repair.

GridGuard transforms complex, fragmented grid data into an actionable, proactive maintenance workflow.

 **RISK**

Deterministic 0–100 composite risk scoring.

 **INSIGHT**

Grounded, explainable AI briefings.

 **PRIORITY**

Objective grid impact ranking.

 **ACTION**

Targeted crew pre-positioning.

GOAL: Transition electrical grid maintenance from reactive firefighting to proactive, data-driven resilience.